# CineHub - Movie Booking Platform

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# 1. Overview

A B2B/B2C movie ticket booking platform built with Java and Spring Boot.

| Users | Operation |
|---|---|
| **Theatre Partners (B2B)** | Register theatres, create and manage shows, control seat inventory, view bookings |
| **Customers (B2C)** | Browse movies by city/genre/language, view show timings, book seats, apply offers, cancel bookings |

# 2 Functional Requirements:

## 2.1 User Management

-Register

-Login

## 2.2 Movie & Show Browsing

-Users can browse movies by city, language, release date

-User can view movie details

-User can view available showtimes for a selected movie in a particular theatre

## 2.3 Theatre & Screen management

- Admin can add/edit/delete theatres, screen and seat map.

## 2.4 Seat selection

-User can view the seat map for a selected showtimes

-Users can select available seats and proceed to booking

## 2.5 Booking & Ticketing

## 2.6 Booking platform offers

- 50% discount applied for third ticket

- Ticket booked for afternoon show get 20% discount

# 3 Nonfunctional Requirements

## 3.1 Performance

-The system should support at least 10000 concurrent users without noticeable latency.

## 3.2 Scalability

System should be horizontally scalable to handle high traffic especially during blockbuster releases.

## 3.3 Availability

System should ensure 99.9% uptime

## 3.4 Data consistency

Booking transactions must ensure ACID properties

## 3.5 Maintainability

Codebase should be modular and easy for maintenance and extension

## 3.6 Localization

The system should support multiple languages

## 3.7 Logging & Monitoring

All operations should be logged for audit and debugging.

System health should be monitored.

# 4 How to run

**Prerequisites**

- Java 20+
- MySQL 8+
- Maven 3.8+
- Spring Boot 3+
- Git
- Docker

Open command prompt; c:

Clone Github project:

git clone https://github.com/dineshmetkari/CineHub

cd CineHub

docker-compose up -d

Open browser & open url:

http://localhost:8080/swagger-ui/index.html

-Register

- Login

-Copy token to Authorize

-Explore API endpoints.

# 5 Technology Stack

| Concern | Choice |
|---|---|
| Language | Java 20 |

| Concern | Choice |
| --- | --- |
| Framework | Spring Boot 3.2.1 |
| Database | MySQL 8 (H2 in-memory for tests) |
| ORM | Spring Data JPA / Hibernate |
| Security | Spring Security + JWT (jjwt 0.12.3) |
| API Docs | SpringDoc OpenAPI 3 (Swagger UI) |
| Build | Maven 3.8+ |
| Connection pool | HikariCP (20 max connections) |
| Cache | Spring Cache (simple in-memory)<br>Assumption: Redis in production |
| Docker | Container |

# 6 Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                        Client Layer                          │
│          (Web browser, Mobile app, Postman / curl)           │
└──────────────────────────────────────────────────────────────┘
                              │
                              ▼
┌──────────────────────────────────────────────────────────────┐
│               Spring Security — JWT Filter Chain             │
│   JwtAuthenticationFilter  →  SecurityContextHolder          │
└──────────────────────────────────────────────────────────────┘
                              │
              ┌───────────────┼───────────────┐
              ▼               ▼               ▼
      ┌──────────────┐ ┌──────────────┐ ┌──────────────┐
      │ AuthController│ │  Customer    │ │  Theatre     │
      │ /api/v1/auth  │ │  Controller  │ │  Controller  │
      │               │ │/api/v1/      │ │ /api/v1/     │
      │               │ │customer      │ │ theatre      │
      └──────────────┘ └──────────────┘ └──────────────┘
              │               │               │
              └───────────────┼───────────────┘
                              ▼
      ┌──────────────────────────────────────────────┐
      │                  Service Layer               │
      ├──────────────────────────────────────────────┤
      │  AuthService       ShowService               │
      │  BookingService    OfferService              │
      │  UserResolverService                         │
      │                                              │
      │  strategy/                                   │
      │    AfternoonShowDiscountStrategy             │
      │    ThirdTicketDiscountStrategy               │
      └──────────────────────────────────────────────┘
                              │
      ┌──────────────────────────────────────────────┐
      │               Repository Layer               │
      ├──────────────────────────────────────────────┤
      │  UserRepository        TheatreRepository     │
      │  MovieRepository       ShowRepository        │
      │  ScreenRepository      SeatRepository        │
      │  BookingRepository     OfferRepository       │
      │  CityRepository                              │
      └──────────────────────────────────────────────┘
                              │
                              ▼
                   ┌──────────────────┐
                   │   MySQL 8        │
                   │  (HikariCP 20)   │
                   └──────────────────┘
```

### Layer responsibilities

**Controller layer** — three controllers, all under `/api/v1`:

| Controller | Prefix | Auth required |
|---|---|---|
| `AuthController` | `/auth` | None (public) |
| `CustomerController` | `/customer` | JWT for booking endpoints; browse/offers are public |
| `TheatreController` | `/theatre` | JWT with `ROLE_THEATRE_PARTNER` on every endpoint |

Controllers extract the authenticated user via `@AuthenticationPrincipal UserDetails` and delegate to `UserResolverService.resolveCurrentUserId()` to get the database `Long` userId. No `?partnerId=` or `?customerId=` query params — identity always comes from the token.

**Service layer** — all business logic, validation, and transaction boundaries live here. Controllers never call repositories directly.

**Repository layer** — one `JpaRepository` per entity, with named JPQL queries for complex lookups. No native SQL; everything goes through the ORM.

## Package structure

```
src/main/java/com/cinehub/booking/
├── config/          # SecurityConfig, OpenApiConfig, AuditConfig, DataSeeder
├── controller/      # AuthController, CustomerController, TheatreController
├── dto/
│   ├── request/     # AuthRequest, BookTicketsRequest, CreateShowRequest, BulkCancelRequest
│   └── response/    # AuthResponse, BookingResponse, ShowDetailsResponse,
│                    # PriceCalculationResponse, SeatResponse
├── entity/          # City, User, Theatre, Screen, Movie, Show, Seat, BookingSeat, Booking, Offer
├── enums/           # UserRole, BookingStatus, SeatStatus, ShowTimeSlot
├── exception/       # ResourceNotFoundException, BookingException,
│                    # InvalidOperationException, GlobalExceptionHandler
├── repository/      # One JpaRepository per entity, with custom JPQL queries
├── security/        # JwtTokenProvider, JwtAuthenticationFilter, UserDetailsServiceImpl
└── service/
    ├── strategy/    # DiscountStrategy (interface), AfternoonShowDiscountStrategy,
    │                # ThirdTicketDiscountStrategy
```

```
    ├── AuthService.java
    ├── BookingService.java
    ├── OfferService.java
    ├── ShowService.java
    └── UserResolverService.java
src/main/resources/
└── application.yml     # MySQL datasource, JWT config, app.offers.* properties, test profile


src/test/java/com/cinehub/booking/
└── BookingWorkflowTest.java
```

# 7 Design patterns

**Strategy Pattern — discount calculation** Offer Service holds a List<DiscountStrategy> injected by Spring. At booking time every strategy's isApplicable() is checked and all matching discounts are **stacked** — both the afternoon and third-ticket discounts can fire together on the same booking. Adding a new offer requires only a new @Component class; no existing code changes.

```
DiscountStrategy (interface)

  ├── AfternoonShowDiscountStrategy  — 20% off shows 12:00–16:00

  └── ThirdTicketDiscountStrategy    — 50% off every 3rd ticket (price-desc sort)
```

**Repository Pattern** All data access is behind JpaRepository interfaces with named JPQL queries. Business logic in services never builds queries inline.

**DTO Pattern** Separate request and response classes for every API surface. Entities never leave the service layer.

# 8 Key design decisions

**1. Concurrency — pessimistic + optimistic locking combined**

**Problem:** Multiple customers booking the same seat simultaneously leads to double-booking.

**Solution — two layers:**

**Layer 1: Pessimistic write lock on seats** (SeatRepository)

```
@Lock(LockModeType.PESSIMISTIC_WRITE)

@Query("SELECT s FROM Seat s WHERE s.seatId IN :seatIds")

List<Seat> findSeatsByIdWithPessimisticLock(@Param("seatIds") List<Long> seatIds);
```

When bookSelectedTickets() begins, a SELECT … FOR UPDATE is issued on the requested seats. Any concurrent transaction requesting the same seat IDs blocks at the database level until the first transaction commits or rolls back. The second transaction then re-evaluates availability and rejects the already-booked seats.

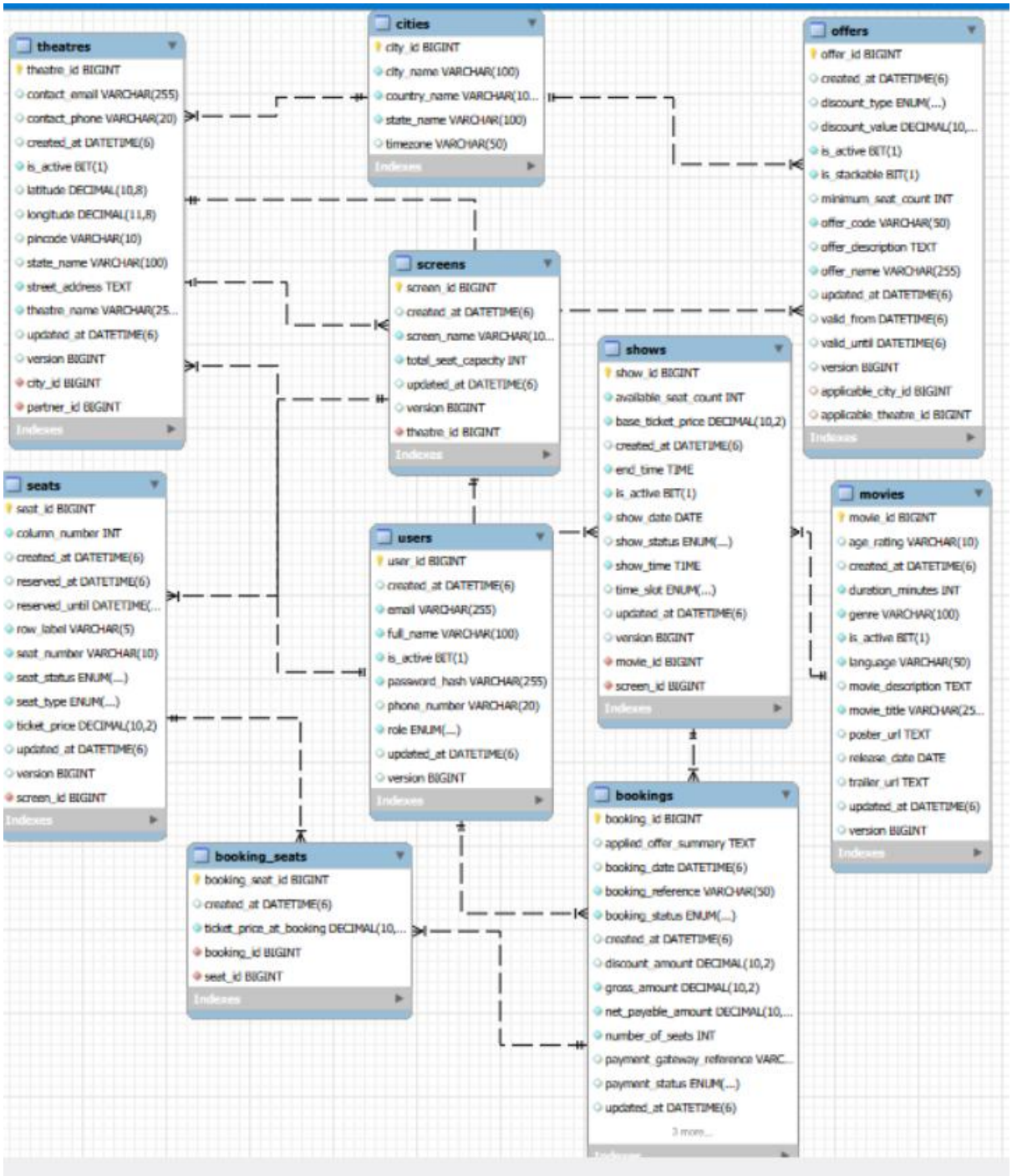
**2. Authentication — stateless JWT**

Every request carries a Bearer token in the Authorization header. JwtAuthenticationFilter intercepts each request before the controller, validates the token via JwtTokenProvider (HS256, jjwt 0.12.3), and populates SecurityContextHolder. Sessions are never created (SessionCreationPolicy.STATELESS).

The partner identity is extracted from the token — no ?partnerId= query param is accepted. UserResolverService.resolveCurrentUserId() converts the email subject from the JWT to the database Long userId on every authenticated call.

# 9 Data model

```
City —< Theatre —< Screen —< Seat
                  |
                 Show >— Movie
                  |
                Booking —< BookingSeat >— Seat
                  |
                 User
```




Key entity decisions:

- **Pessimistic write lock** (PESSIMISTIC_WRITE) on the seat query inside BookingService prevents two concurrent transactions from booking the same seat simultaneously.

# 10 Getting started - Configuration

**1 — Create the MySQL database**

CREATE DATABASE movie_booking CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

**2 — Configure credentials**

Edit src/main/resources/application.yml:

```
spring:
  datasource:
    url: jdbc:mysql://localhost:3306/movie_booking?useSSL=false&serverTimezone=UTC&allowPublicKeyRetrieval=true
    username: your_username
    password: your_password
```

**3 — Configure the JWT secret**

Replace the default secret with a strong random string of at least 32 characters:

```
jwt:
  secret: your-strong-secret-key-at-least-32-characters
  expiration-ms: 86400000   # 24 hours
```

**4 — Build and run**

```
mvn clean install
mvn spring-boot:run
```

The application starts on http://localhost:8080.
On first startup DataSeeder runs automatically and populates the database with demo cities, users, theatres, screens, seats, movies, and shows.

**5 — Swagger UI**

http://localhost:8080/swagger-ui.html

Use **Authorize** (top right) and paste the JWT token from the login response as Bearer <token>.

---

**Demo seed data**

DataSeeder creates the following on first startup (idempotent — skipped if data exists):

| Type | Details |
|---|---|
| Cities | Mumbai, Bangalore, Delhi |
| Users | customer@example.com, partner@example.com, admin@example.com (password: Password@123) |
| Theatres | PVR Cinemas — Lower Parel (Mumbai), INOX — Koramangala (Bangalore) |
| Screens | 2 screens per theatre (Screen 1: 100 seats, Screen 2: 80 seats) |
| Seats | Rows A–C → PREMIUM (₹300), Rows D–J → REGULAR (₹200) |
| Movies | The Great Adventure (Hindi/Action), Comedy Nights (Hindi/Comedy), The Silent Thriller (English/Thriller) |
| Shows | Morning (10:00), **Afternoon (14:00)**, Evening (18:00), Night (21:30) — today and tomorrow |

The 14:00 show is in the afternoon window so both discount strategies fire immediately when test with 3+ seats.

# 11 API reference

## a. Authentication — /api/v1/auth

| Method | Path | Body | Description |
|---|---|---|---|
| POST | /register | { fullName, email, password, phoneNumber, role } | Register (role: CUSTOMER or THEATRE_PARTNER) |
| POST | /login | { email, password } | Returns { accessToken, tokenType, userId, fullName, role } |

## b. Customer — /api/v1/customer

All browse endpoints are public. Booking endpoints require Authorization: Bearer <token>.

| Method | Path | Query params | Description |
|---|---|---|---|
| GET | /movies | city, language, genre (all optional) | Browse active movies |
| GET | /movies/{movieId}/shows | city, date | **Read Scenario 1** — theatres and show timings for a movie in a city on a date |
| GET | /shows/{showId}/seats | — | Available seats for a show (seat grid) |

| Method | Path | Query params | Description |
|---|---|---|---|
| POST | /offers/calculate | showId, seatIds | **Read Scenario 2** — preview pricing with all applicable discounts |
| POST | /bookings | — | **Write Scenario 1** — book tickets (discounts applied automatically) |
| POST | /bookings/bulk | — | Bulk book — many seats for one show in one transaction |
| GET | /bookings | — | My booking history |
| GET | /bookings/{reference} | — | Booking detail by reference number |
| DELETE | /bookings/{reference} | — | Cancel a booking |
| DELETE | /bookings/bulk | — | Bulk cancel |

**Book tickets request body:**

```
{
  "showId": 1,
  "selectedSeatIds": [10, 11, 12],
  "promoCode": "optional"
```

}

## c. Theatre partner — /api/v1/theatre

All endpoints require Authorization: Bearer <token> with role THEATRE_PARTNER.

| Method | Path | Description |
|---|---|---|
| POST | /shows | **Write Scenario 2** — create a show (checks for scheduling conflicts) |
| PUT | /shows/{showId} | Update show date, time, or price (blocked if bookings exist) |
| DELETE | /shows/{showId} | Cancel a show (blocked if bookings exist) |
| GET | /shows | View shows at a theatre on a date (?theatreId=&date=) |
| GET | /shows/{showId}/seats | View full seat layout and statuses |
| PUT | /shows/{showId}/seats | **Write Scenario 3** — bulk update seat status (?seatIds=&newStatus=BLOCKED) |
| GET | /shows/{showId}/bookings | View all bookings for a show |

**Create show request body:**

```
{
  "movieId": 1,
  "screenId": 1,
  "showDate": "2026-03-20",
  "showTime": "14:30",
  "baseTicketPrice": 250.00
}
```

---

**Offer implementation**

Both offers are configured in application.yml and applied automatically — no promo code required:

```
app:
  offers:
    third-ticket-discount-percent: 50
    afternoon-discount-percent: 20
    afternoon-start-hour: 12
    afternoon-end-hour: 16
```

**Third ticket discount (50% off every 3rd seat)** Seats are sorted by price descending before applying the discount so the highest-priced 3rd seat always gets discounted.

Example — 5 seats [₹300, ₹300, ₹300, ₹200, ₹200] sorted desc:

- Position 3: ₹300 → 50% off = **₹150 discount**
- Position 6: does not exist
- Total discount: ₹150

**Afternoon show discount (20% off total)** Applies when showTime >= afternoonStartHour and showTime < afternoonEndHour.

**Stacking** Both discounts fire independently and are **added together**. A 3-ticket booking at 14:00 gets both:

Gross: ₹900 (3 × ₹300)

3rd seat: −₹150 (50% of ₹300)

Afternoon:−₹180 (20% of ₹900)

Net: ₹570

# 12 Configuration reference

| Property | Default | Description |
|---|---|---|
| jwt.secret | (set this) | HS256 signing key — minimum 32 characters |
| jwt.expiration-ms | 86400000 | Token lifetime (24 hours) |
| app.offers.third-ticket-discount-percent | 50 | % off every 3rd ticket |
| app.offers.afternoon-discount-percent | 20 | % off afternoon shows |
| app.offers.afternoon-start-hour | 12 | Afternoon window start (inclusive) |
| app.offers.afternoon-end-hour | 16 | Afternoon window end (exclusive) |
| spring.jpa.hibernate.ddl-auto | update | Use validate in production with Flyway |

# 13 Future enhancements

- Payment gateway integration (Razorpay / Stripe)
- Email and SMS booking confirmations
- Seat hold with expiry timer (reserve for 10 minutes before payment)
- Redis cache for movie catalogue and show schedules
- Analytics dashboard for theatre partners
- Microservices and Kafka implementation
- Deployment on EKS
- Terraform for DevOps